

SEEDLINE · CALCULATION NOTE

SeedLine sizing and first-principles checks

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (drive and spacing, metering, plates, mass, push force, slip, structure, work rate, depth, uniformity, cost) against every requirement

SeedLine sizing and first-principles checks

On paper the seeder works as a ground-driven single-row planter: the drive covers 22 to 589 mm of in-row spacing with the optional ratio kit, slip from the metering load is under 2 %, the parts are lightly stressed and the work rate is about 0.142 ha/h. Version 0.2 applies the recommendations Amish accepted on 2026-09-25 (SDL-DDR-002): a lighter 22 x 1.2 mm handle tube, the 12 and 18 T wheel sprockets moved to an optional ratio kit, a design-case walking speed of 2.9 km/h (the walking-speed rule for R1), and R8 accepted for field crops on ploughed land. Thirteen of the eighteen requirements are now met on paper, R11 and R17 with thin margins (14.9 kg with the marker kit against 15 kg; \$198 base against about \$200), and two cannot be verified at TRL 3. Three are **at risk**: R1 (cell speed 0.299 m/s at 2.9 km/h against an assumed 0.30 m/s limit, no margin), R2 (spacing CV about 31 % with assumed miss, double and scatter rates) and R10 (132 N push in the design case, but 177 N at best on a heavy, loose seedbed). None is not met.

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the key dimensions, sprocket geometry, cell sizes, hopper volume and handle geometry from the parametric model `cad/src/model.py`, and the prices from `bom/bom.csv`, so the model, the drawing SDL-DWG-001, the BOM and this note agree. All values are first-principles estimates; nothing here is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design unless marked as decided.

Input	Value	Basis
Architecture	Vertical cell plate, #35 chain, front drive wheel, rear press wheel, bolted frame, marker arm; 15 T wheel sprocket on the base seeder, 12 and 18 T in an optional ratio kit; 22 x 1.2 mm handle tube	Decided, SDL-DDR-001 items 1, 3, 4, 7 and 8; SDL-DDR-002 items 3 and 4
Design case	Maize at 250 mm on 0.75 m rows, 50 mm deep, 2.9 km/h (0.81 m/s), the walking-speed rule for R1	SDL-REQ-001 v0.4; SDL-DDR-002 item 5 (was 3 km/h)
Drive wheel	300 mm over the lug tips, 45 mm wide, 10 mm lugs; travel 942.5 mm per turn before slip	Model parameter
Typical slip for spacing tables	5 %	Allowance; the calculated skid (section 6) is lower
Cell fill limit	0.30 m/s cell speed	Assumed working limit for a gravity-filled cell plate; to be checked by bench test at TRL 4
Cell size	Length $1.15 L + 0.5$ mm, depth $1.05 W + 0.3$ mm, plate thickness the larger of 6 mm and $1.2 T$ (L, W, T: seed length, width, thickness)	Rule used by <code>make_plate</code> in the model
Seed dimensions	Maize 12 x 8 x 5 mm, sorghum 4.5 x 4 x 3, groundnut kernel 15 x 9 x 8, common bean 13 x 8 x 6, spinach 3.5 x 3 x 2.5, pelleted carrot 3.3 mm, raw carrot 3 x 1.2 x 0.6	Typical values; varieties differ
Print	± 0.2 mm on a 0.4 mm nozzle, tolerance limited to 10 % of the smallest cell dimension; 1.6 mm of skins, 1.2 mm walls, 25 % infill; 6 mm ³ /s average rate; PETG 1.27 g/cm ³ at \$25/kg	Common desktop FDM printer
Meter torque	1.0 N·m at the plate shaft (design value)	First-principles estimate is 0.20 N·m (section 3); the design value is five times higher
Soil	Cone index 300, 200 and 100 kPa for good, design and heavy (loose, cloddy) seedbeds; opener specific resistance 20, 30 and 40 kPa; covering chains and press 10, 15 and 20 N	Assumed; no soil data from a partner region yet
Rolling resistance	Brixius motion resistance, $MR/W = 1/B_n + 0.04$, with $B_n = CI b d / W / (1 + 3 b/d)$ for a rigid wheel	Brixius (1987), an empirical model fitted to larger tires; used here outside its calibrated range
Opener draft	Specific resistance x 2.5 x 18 mm runner width x depth	Narrow-tool failure zone taken as 2.5 tool widths
Field work	50 m rows, 15 s per turn, 90 s per refill, 85 % working time	Assumed for a small plot and one operator
Structure	S235 class steel, yield 235 MPa; dynamic factor 3 for drops and stones; 100 N side load at the grip	Assumed
Uniformity model	5 % misses, 4 % doubles (second seed 0 to 30 mm behind), 15 mm drop scatter, slip 5 ± 1.5 %	The miss and double rates are the TRL 2 estimates; not calculable from first principles

■ 2. Drive, spacing and range (R3, R4)

The spacing is $s = \pi D / (n i) / (1 - \sigma)$, with $\pi D = 942.5$ mm, n plate cells, ratio $i = \text{wheel teeth} / 15$ and travel reduction σ . The three wheel sprockets give $i = 0.8, 1.0$ and 1.2 (pitch diameters 36.80, 45.81 and 54.85 mm; plate sprocket 15 T). The base seeder carries the 15 T wheel sprocket ($i = 1.0$); the 12 and 18 T sprockets are an optional ratio kit (SDL-DDR-002 item 4).

Table 2. Example plates. Nominal spacing, and with the 5 % slip allowance.

Crop	Target (mm)	Cells	Ratio	Nominal (mm)	With 5 % slip (mm)
Maize (design case)	250	4	1.0	235.6	248.0
Sorghum, groundnut	150	6	1.0	157.1	165.3
Common bean	100	9	1.0	104.7	110.2
Spinach	50	16	1.2	49.1	51.7
Carrot, lettuce (pelleted)	25	30	1.2	26.2	27.6
Skip-cell plate	400	2	1.2	392.7	413.4
Skip-cell plate	450	2	1.0	471.2	496.0
Skip-cell plate	600	2	0.8	589.0	620.1
Shortest		36	1.2	21.8	23.0

With skip-cell plates (one or two cells; decided, SDL-DDR-001 item 5) and the ratio kit the nominal range is **21.8 to 589 mm**, and 1,178 mm with a single cell, so **R3 (25 to 400 mm) is met**. The settings are discrete: choosing the best plate and ratio for every target from 25 to 400 mm, the worst error is 11.1 % (a 372 mm target falls between 331 mm and 413 mm). The spacing card should list the nearest setting for each crop. The base seeder alone (15 T only) covers 26.2 to 471 mm nominal, but its settings are coarser at long spacings: the worst error is 17.3 % (a 400 mm target falls between 331 mm and 496 mm). The ratio kit is needed for the 25 mm pelleted-seed plate at 1.2 and for close settings above about 300 mm.

Chain. The sprocket centers are 286.2 mm apart. A #35 chain needs 73.61, 75.10 and 76.61 pitches for the 12, 15 and 18 T wheel sprockets, so the chain is set to 74, 76 or 77 links (the 77-link length needs one offset link). The largest slack left over is 8.6 mm (15 T, 76 links), which a spring idler takes up by deflecting the slack strand by about 35 mm. The idler in `bom/bom.csv` has 45 mm of travel.

■ 3. Metering speed and cell fill (R1)

Amish accepted the walking-speed rule on 2026-09-25 (SDL-DDR-002 item 5): the operator walks at about 2.9 km/h with the 0.8 and 1.0 ratios and about 2.4 km/h with the 1.2 ratio. The design case now uses 2.9 km/h (it was 3 km/h in v0.1). At 2.9 km/h the wheel turns at 51.3 rpm. For the maize plate the seed centers sit on a 111.3 mm circle, so:

Table 3. Plate and cell speeds at 2.9 km/h.

Ratio	Plate speed (rpm)	Cell speed (m/s)	Walking speed for 0.30 m/s (km/h)
0.8	41.0	0.239	3.64
1.0	51.3	0.299	2.91
1.2	61.5	0.359	2.43

The design case runs at 0.299 m/s, at the assumed 0.30 m/s limit with no margin, and the limit itself is an assumption, so **R1 stays at risk**. A simple kinematic indicator (a still seed must fall the cell depth while the open cell passes it) gives 0.20 m/s for the maize cell; seeds in the pool are dragged along by the plate, so real plates fill at higher speeds, but the indicator confirms that cell speed, not the drive, sets the walking speed. The maize plate passes 3.4 cells per second; the 36-cell plate at 1.2 passes 36.9 cells per second at 2.9 km/h, which is demanding for the brush (the rule of 2.4 km/h at 1.2 lowers that further). Misses and multiples depend on seed shape, brush setting and cell speed and cannot be calculated on paper; they need the bench test that TRL 4 would bring.

Meter torque. Friction of the seed pool on the lower quarter of both plate faces (Janssen-type wall pressure under a 0.2 m seed head) is about 0.051 N·m, the brush about 0.048 N·m and the bearings and chain about 0.10 N·m, a total of **0.20 N·m**. The calculations below use 1.0 N·m, five times this estimate, to cover seed jamming at the brush.

A smaller plate would lower the cell speed in proportion to the cell circle. Under SDL-DDR-002 item 5 the plate size is revisited only with bench data, which is TRL 4 work and on hold.

■ 4. Plates per crop (R5, R6)

Table 4. Cell sizes, print tolerance, mass and print time for each starter plate.

Plate	Cell length x depth (mm)	Plate thickness (mm)	Cells (maximum)	Tolerance / smallest cell dimension	Mass (g PETG)	Print time (h)
Maize	14.3 x 8.7	6	4 (21)	2.3 %	40.0	1.46
Sorghum	5.7 x 4.5	6	6 (47)	4.4 %	41.1	1.50
Groundnut kernel	17.8 x 9.8	10	6 (17)	2.1 %	53.8	1.96
Common bean	15.4 x 8.7	8	9 (20)	2.3 %	45.8	1.67
Spinach	4.5 x 3.5	6	16 (56)	5.8 %	41.1	1.50
Carrot, pelleted	4.3 x 3.8	6	30 (58)	5.3 %	41.1	1.50
Carrot, raw	3.9 x 1.6	6	30 (62)	12.8 %, too fine	41.2	1.50

Raw seed from 3.5 mm (spinach) to 15 mm (groundnut) fits the tolerance rule. Raw carrot seed needs a 1.6 mm deep cell, where ± 0.2 mm is 12.8 % of the cell, so 1 to 2 mm seed must be sown pelleted (about 3.3 mm pellets), as decided (SDL-DDR-001 item 6). With that redefinition **R5 is met** on paper. Every starter plate prints in 2 h or less, but the 10 mm groundnut plate takes 1.96 h, so **R6 is met with almost no margin** for thick plates; the ± 0.2 mm cell accuracy cannot be verified at TRL 3. The TRL 2 estimate of about 25 g and 1 h per plate was low; the model gives 40 to 54 g and 1.5 to 2.0 h.

■ 5. Mass and center of mass (R11)

Table 5. Mass by group.

Group	Mass (kg)	Notes
Wheels and axles (items 1, 11)	3.94	2.2 kg drive wheel, 1.2 kg press wheel, two 16 mm axles
Drive and metering (items 2 to 8)	2.00	Hopper from the model volume, 3 mm PETG walls
Opener and covering (items 9, 10)	1.59	
Frame and handle (items 12, 13)	5.97	Frame 3.63 kg (1.65 m of 25 x 25 x 1.5 tube, drops, posts, brackets); handle 2.34 kg (two 978 mm side tubes of 22 x 1.2 round tube, grip, brace, sleeves)
Hardware (item 15)	0.45	
Base seeder	13.94	
Row marker kit (item 14)	1.00	
With marker kit	14.94	

In v0.1 the handle was 25 x 1.5 mm round tube (2.99 kg), which put the seeder at 14.6 kg base and 15.6 kg with the marker kit, so R11 was not met. Amish accepted the lighter handle on 2026-09-25 (SDL-DDR-002 item 3): 22 x 1.2 mm round tube saves 0.65 kg, so the base seeder is 13.9 kg and the seeder with the marker kit is 14.9 kg. **R11 is now met, with only 0.06 kg of margin** with the kit fitted. The side tube's factor on yield under the 100 N side load falls from 2.9 to 1.9 (section 7). The center of mass of the base seeder is 358 mm behind the drive axle and 245 mm above the ground; a full hopper adds 1.73 kg of maize. The rails were sized down from 2 mm to 1.5 mm wall in v0.1 because the stresses are low.

■ 6. Push force and slip (R10, R4)

The push is found from the planar statics of the seeder on its two wheels, with the operator's force at the grip (950 mm high, 647 mm behind the press wheel) at an angle ϕ below the horizontal. Rolling resistance acts at ground level and the opener draft 25 mm below it. For each case the script finds the smallest horizontal push that balances the resistance with both wheels on the ground.

Table 6. Push force (horizontal component) with a full hopper, 154 N total weight.

Seedbed	Along the handle (50°)	Lowest push, at angle	Angles that balance	Rolling / opener / chains (N)	Wheel loads, drive / press (N)
Good	80 N	71 N at 15°	11 to 50°	25 / 45 / 10	88 / 161
Design	132 N	108 N at 30°	22 to 50°	49 / 68 / 15	105 / 206
Heavy, loose	no balance	177 N at 36°	30 to 48°	67 / 90 / 20	212 / 71 (at 36°)

Three findings follow. First, a purely horizontal push has no balance in any case: at grip height it pitches the seeder forward and lifts the press wheel, so the TRL 2 comparison of a horizontal push against a push along the handle does not apply. Second, in the design case the push along the handle is 132 N, inside the 150 N target with a 12 % margin that rests on assumed soil values (the lighter handle took about 4 N off the v0.1 figure of 136 N). Third, on a heavy, loose seedbed a push along the handle drives the press wheel into the soil faster than it moves the seeder (no balance), and the best push is 177 N at 36°. **R10 is at risk.** The equivalent rolling coefficients are 0.102, 0.159 and 0.236, inside the 0.2 to 0.3 range assumed at TRL 2 only for the heavy case.

Slip. The drive wheel must supply the meter torque through the chain: 6.7 N at the rim at a 1.0 ratio and 8.0 N at 1.2. Against 88 to 212 N of wheel load, the Brixius traction relation gives a skid of **1.2 to 1.7 %**, well inside the 8 % of R4. The larger uncertainty is the rolling radius: if the lugs sink fully, the wheel rolls on its rim and travels up to 6.7 % less per turn, which makes the spacing shorter, not longer. The two effects partly cancel. **R4 is met on paper;** the spacing card should be calibrated by counting wheel turns over a measured 10 m on the user's own seedbed.

■ 7. Structure

Table 7. Stress checks.

Part	Load case	Result
Side rails, 25 x 25 x 1.5 tube	Heavy case wheel and push loads, factor 3, plus opener clamp moment	69 MPa, factor 3.4 on yield
Handle side tube, 22 x 1.2 round (SDL-DDR-002)	100 N side load at the grip	126 MPa, factor 1.9 (was 80 MPa, factor 2.9, with 25 x 1.5 tube)
Press axle, 16 mm	Heavy case load, factor 3	19 MPa
#35 chain	1.2 N·m on the 15 T plate sprocket	52 N against about 7.8 kN minimum tensile strength (typical catalog value)
Plate hub D-flat	1.2 N·m in PETG	5.7 MPa bearing stress

The frame and drive are lightly stressed. The lighter handle is the most highly stressed steel part, still with a factor of 1.9 on yield under a static side load; fatigue of the telescoping joint is a TRL 4 question. Durability (R15) depends on wear of the plates, brush, chain and opener, which cannot be calculated at TRL 3.

■ 8. Work rate and hopper (R9, R12)

The theoretical capacity on 0.75 m rows at the 2.9 km/h design walking speed is 0.217 ha/h. With 50 m rows, 15 s turns, 90 s refills and 85 % working time the field efficiency is 65 %, so the field capacity is **0.142 ha/h (7.1 h/ha)**, which meets R9 and compares with about 56 h/ha for hoe planting (SDL-PRB-001). The walking-speed rule costs about 0.003 ha/h against the v0.1 figure at 3 km/h (0.145 ha/h). On 0.3 m vegetable rows the capacity is 0.057 ha/h.

The hopper cavity in the model is **2.40 L** (R12 met). It holds 1.73 kg of maize, about 5,754 seeds, enough for 1,438 m of row or 0.108 ha per fill. Maize at 0.75 x 0.25 m needs 53,333 seeds per hectare.

■ 9. Depth control (R8)

The depth bracket gives 10 to 60 mm in 10 mm steps (six holes in the shank). The opener point is 182 mm behind the drive axle, so the frame, riding on its two wheels, raises the opener by 0.69 of any rise under the drive wheel and 0.31 of any rise under the press wheel. Holding ± 10 mm therefore needs the wheel path to stay within about ± 15 mm under the drive wheel or ± 32 mm under the press wheel. A 50 mm clod under the drive wheel lifts the opener by 34 mm. Amish accepted the recommendation on 2026-09-25 (SDL-DDR-002 item 6): the rigid opener is accepted for field crops on ploughed land, and R8 is restated for a ploughed field-crop seedbed, where the ± 15 mm wheel-path condition is expected to hold. On that basis **R8 is met on paper**; on rough, hand-tilled seedbeds depth will vary more, and a spring-loaded opener or depth-gauge shoe is to be reviewed with partner data.

■ 10. Spacing uniformity (R2)

A Monte Carlo run of 20,000 cells for the maize plate, with the assumed rates in Table 1, gives a coefficient of variation of **30.7 %** for all plant-to-plant spacings, against a target of 30 % or less. The precision of the single spacings (those between 0.5 and 1.5 times the mean) is 8.9 %, with a miss index of 4.8 % and a multiple index of 3.9 %. The all-spacings CV is driven almost entirely by the misses and doubles, not by drop scatter or slip. **R2 is at risk**; it hinges on miss and double rates that only a bench test can give.

■ 11. Cost (R16, R17)

All 16 BOM lines are priced. The base seeder (items 1 to 13 and 15) costs **\$197.50**, the marker kit \$24.00 and the optional ratio kit (item 16: 12 and 18 T sprockets and an offset link) \$16.00, a total of **\$237.50** against the \$400 budget in `project.yaml` (R16 met). In v0.1 the base seeder carried all three wheel sprockets and cost \$213.50, so R17 was not met. Moving the 12 and 18 T sprockets to the optional ratio kit and the lighter handle tube (SDL-DDR-002 items 3 and 4) bring the base to \$197.50, 1.2 % under the about \$200 target, so **R17 is met with a thin margin**.

■ 12. Results against requirements

Table 8. Every requirement in SDL-REQ-001 v0.4, at-risk items first (none is not met).

ID	Requirement	Value (this note)	Target	Status
R1	Single-seed placement	Cell speed 0.299 m/s at the 2.9 km/h walking-speed rule against an assumed 0.30 m/s limit, no margin	Misses and multiples 5 % or less each	At risk
R2	Spacing uniformity	CV 31 % of all spacings; 9 % for singles (assumed rates)	CV 30 % or less	At risk
R10	Push effort	132 N along the handle (design case); 177 N at best on a heavy, loose seedbed	150 N or less	At risk
R7	Crop change	Side door and hand knob in the model	2 min or less, no tools	Not verifiable at TRL 3
R15	Durability	Stresses low (section 7); wear life unknown	5 seasons; plates 1 season	Not verifiable at TRL 3
R11	Mass and handling	13.9 kg base; 14.9 kg with the marker kit	15 kg or less	Met, 0.06 kg margin
R17	Replication cost	\$197.50 base seeder	About \$200 or less	Met, 1.2 % margin
R8	Sowing depth	10 to 60 mm in 10 mm steps; ± 10 mm for wheel-path bumps within about ± 15 mm	± 10 mm on a ploughed field-crop seedbed (restated)	Met on paper
R3	Spacing range	22 to 589 mm nominal with the ratio kit, every 25 to 400 mm target within 11.1 %; base seeder 26 to 471 mm, within 17.3 %	25 to 400 mm	Met (with the ratio kit)
R4	Spacing follows travel	Skid 1.2 to 1.7 %; rolling radius uncertain by up to 6.7 %	Slip 8 % or less	Met
R5	Seed size range	3.5 to 15 mm raw seed; 1 to 2 mm seed as pellets of about 3.3 mm	1.5 to 15 mm, seed under 2 mm pelleted	Met
R6	Printable plates	1.46 h and 40 g (maize); up to 1.96 h (groundnut); accuracy not verifiable	2 h or less	Met, no margin for thick plates
R9	Work rate	0.142 ha/h (7.1 h/ha)	0.1 ha/h or more	Met
R12	Hopper	2.40 L; 1,438 m of maize row per fill	2 L or more	Met
R13	Row-spacing kit	Marker reach 200 to 900 mm, set at 750 mm in the model	200 to 900 mm, ± 25 mm	Met (design review)
R14	Guarding	Band guard with an outboard face plate over both sprockets and the idler	Nip points covered	Met (design review)
R16	Prototype cost	\$238	\$400 or less	Met
R18	Local build	Bolted 25 mm square tube, 6 mm plate, #35 chain	Common sections, drill and bolts	Met (design review)

Summary: 13 met, 3 at risk, 0 not met, 2 not verifiable at TRL 3 (v0.1: 10 met, 4 at risk, 2 not met, 2 not verifiable).

■ 13. Corrections to the TRL 2 figures

Quantity	TRL 2 (SDL-PRC-001 v0.2)	TRL 3 (v0.1 of this note; see Table 14 for v0.2)
Spacing range	About 22 to 393 mm; R3 not met	21.8 to 589 mm with skip-cell plates; R3 met
Plate mass and print time	About 25 g, about 1 h	40 to 54 g, 1.5 to 2.0 h
Hopper	About 2.1 L, about 1,250 m per fill	2.40 L, 1,438 m per fill
Mass	13.1 kg base, 14.1 kg with marker	14.6 kg base, 15.6 kg with marker
Push force	About 100 N good, about 220 N heavy	83 N good, 136 N design, 178 N heavy at best
Slip	3 to 8 %	1.2 to 1.7 % skid, plus up to 6.7 % rolling radius uncertainty
Work rate	About 0.14 ha/h	0.145 ha/h
Parts cost	\$199 base, \$223 with marker	\$214 base, \$238 with marker
Meter torque	0.5 to 1.0 N·m	0.20 N·m estimated; 1.0 N·m used

Table 14. Changes in v0.2 from the recommendations accepted by Amish (SDL-DDR-002).

Quantity	v0.1	v0.2
Handle tube	25 x 1.5 mm, 2.99 kg handle	22 x 1.2 mm, 2.34 kg handle
Mass	14.6 kg base, 15.6 kg with marker (R11 not met)	13.9 kg base, 14.9 kg with marker (R11 met)
Base parts cost	\$213.50 (R17 not met)	\$197.50 (R17 met); ratio kit \$16.00 optional
Design walking speed	3 km/h, cell speed 0.309 m/s	2.9 km/h, cell speed 0.299 m/s
Work rate	0.145 ha/h	0.142 ha/h
Push, design case	136 N along the handle	132 N along the handle
Push, heavy seedbed	178 N at 35°	177 N at 36°
Handle side tube	Factor 2.9 on yield	Factor 1.9 on yield
R8	At risk	Restated for ploughed field-crop seedbeds; met on paper

Safety: These figures are paper estimates. The chain drive, opener point and wheel lugs remain hazards whatever the numbers say; the guard must be fitted in use and plate changes done with the wheel held still (SDL-PRC-001, Safety). Treated seed is toxic; follow the seed label.